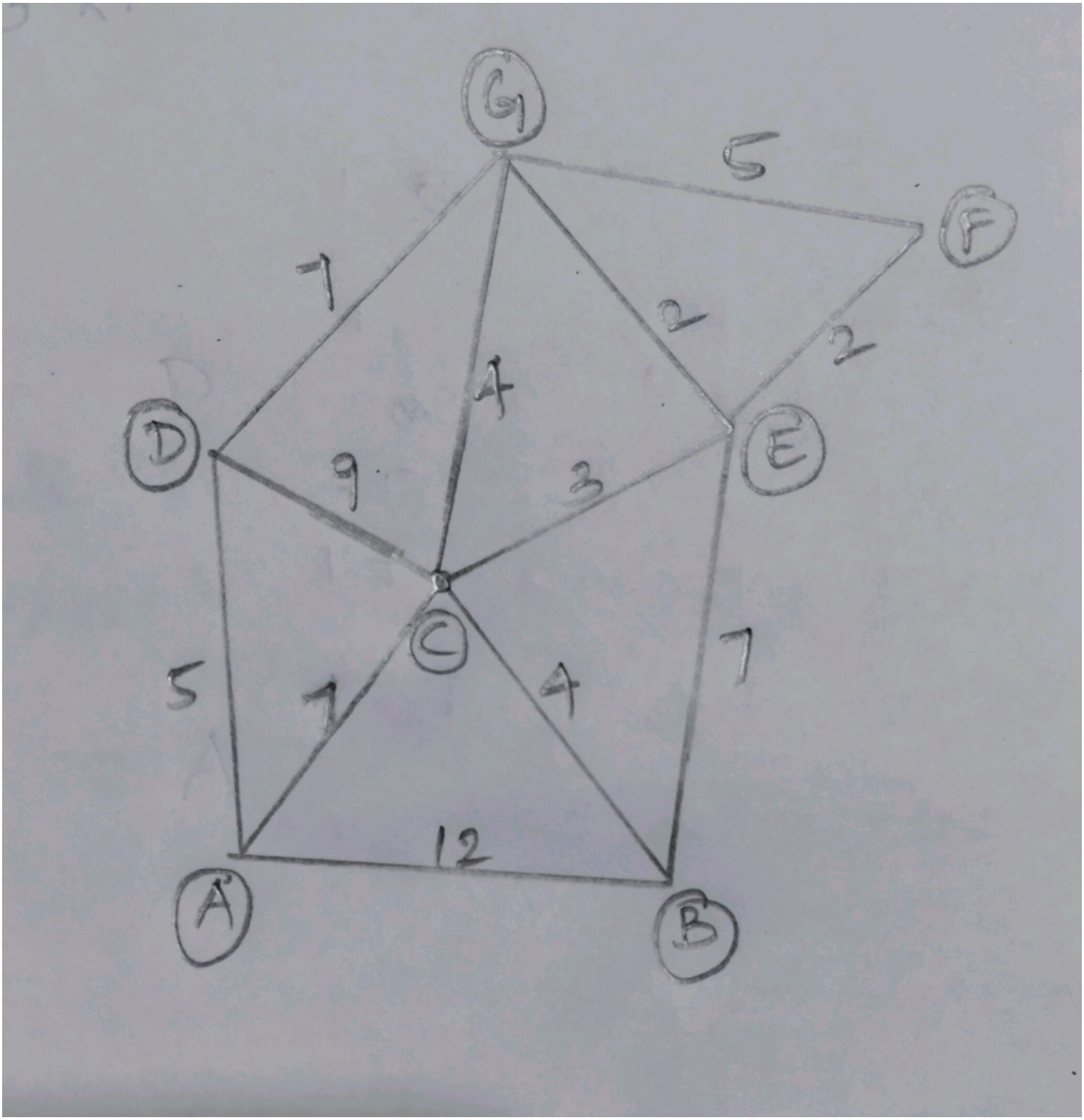

G
5
F
7
2
2
4
D
E
9
3
C
5
7
4
7
12
A
B

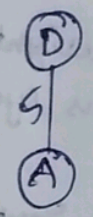
Prim's algorithm:-

Prim's algorithm grows the MST from a starting vertex always adding the cheapest edge that connects a vertex in the tree to a vertex outside the tree. Starting at vertex A.

⇒ let us start with vertex A.

edges ⇒ (A, D) = 5; (A, C) = 7; (A, B) = 12

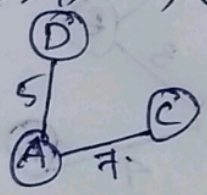
move on with (A, D) = 5



⇒ Available edges (A, C) = 7, (A, B) = 12

(D, G) = 7; (D, C) = 9

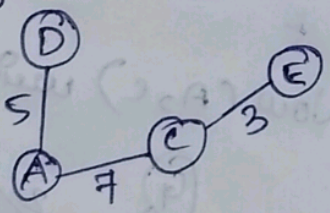
we go with (A, C).



⇒ Available edges (A, B) = 12; (D, G) = 7

(C, B) = 4; (C, E) = 3; (C, G) = 4.

we go with (C, E) = 3.

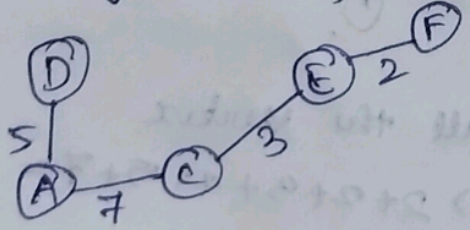


⇒ Available edges (A, B) = 12; (D, G) = 7;

(C, B) = 4; (C, G) = 4; (E, F) = 2; (E, G) = 2;

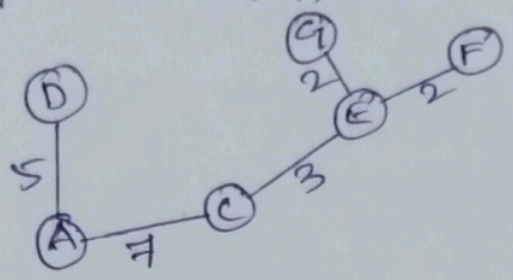
(E, B) = 7.

we go with (E, F) = 2.



⇒ Available edges (A, B) = 12; (D, G) = 7; (C, B) = 4; (C, G) = 4,
(E, G) = 2; (E, B) = 7, (F, G) = 5

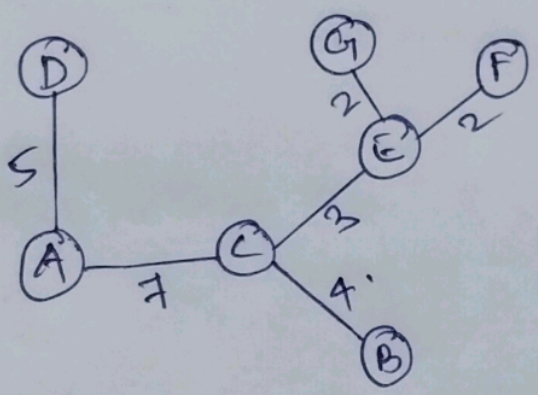
we go with (E, G) = 2.



⇒ Available edges (A, B) = 12; (D, G) = 7; (C, B) = 4;
(C, G) = 4; (E, B) = 7; (F, G) = 5

here (C, G), (F, G), (D, G) form cycle.

So we move on with (C, B) = 4.



Now we have connected all the vertices.

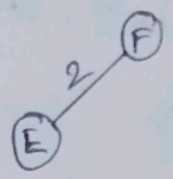
So, the total weight of the graph is

5 + 7 + 3 + 2 + 2 + 4 ⇒ 23.

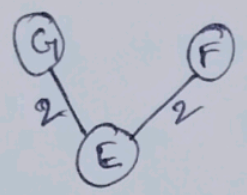
Kruskal's algorithm:

→ It is a greedy approach that sorts all edges by weight and adds them to the MST if they do not form a cycle.

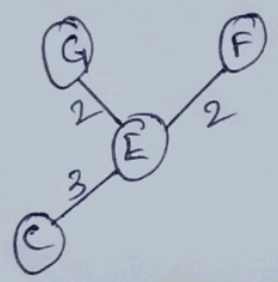
we move on with (E, F) weight = 2.



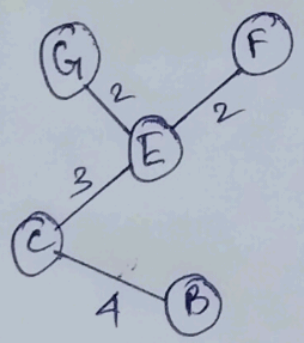
⇒ Now (E, G) weight = 2.



⇒ Now (C, E) weight = 3.

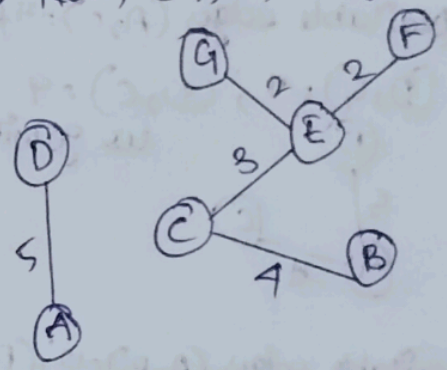


⇒ Now (B, C) weight = 4



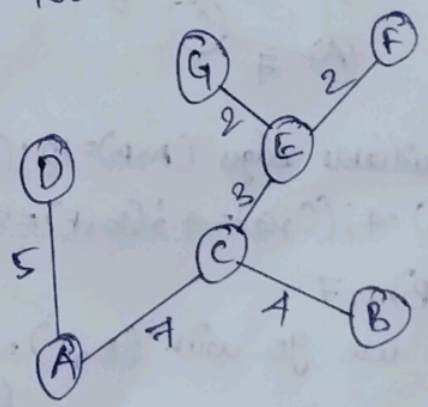
⇒ Now (C, G) weight = 4
we can't (It forms cycle)

⇒ Now, (A, D) weight = 5



⇒ Now, (F, G) weight = 5
(we can't because cycle forms)

⇒ Now (A, C) weight = 7.



So now we have covered all the vertex
the total weight of the graph is ⇒ 2+2+3+4+5+7 = 23